



Unit 4 · Outcome 2 · Energy Sources

# The Two Laws That Decide Every Energy Choice

**KK06** VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



## Why a physics rule belongs in Environmental Science

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01 Why a physics rule belongs in Environmental Science The study design asks you to explain the implications of the first and second laws of thermodynamics when making energy choices . Not the maths of efficiency (that was KK05) — the consequences . These two laws are the reason a coal plant throws away two-thirds of its fuel, the reason "free energy" machines are scams, and the reason cutting waste beats building more supply. The Tarra Cove decision. The town of Tarra Cove, on the edge of Bass Strait, has to power its new desalination plant. The council

## The two laws, stated cleanly

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02 The two laws, stated cleanly Learn both the formal statement (for the high-mark questions) and the plain-English version (so you actually understand it). First Law · Conservation Energy is never created or destroyed "Energy cannot be created or destroyed, only transformed from one form into another. The total energy of an isolated system is constant." Whatever energy you start with, you still have — it has just changed form . If only 33 units come out useful, the other 67 didn't vanish; they became low-grade heat spread into the surroundings. Bookkeep

## First law: the energy is still there

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03 First law: the energy is still there conservation The most common misread is thinking inefficiency means energy "lost". It isn't lost — it's relocated and reshaped . When the Wirindook turbine is 40% efficient, the missing 60% is now warm cooling water in the Kalduke River, warm flue gas up the stack, and friction heat in the bearings. Add it all up and you're back to 100%. Analogy · the energy bank account Think of energy like money in a sealed economy. You can move it between accounts (chemical → heat → motion → electricity) but you can never print

## Second law: energy quality only ever falls

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04 Second law: energy quality only ever falls entropy ↑ The first law treats every joule as equal. The second law says they're not. High-grade energy is concentrated and ordered — it can do work (electricity, the chemical energy in petrol, a tank of compressed steam). Low-grade energy is dispersed and lukewarm — heat spread thinly through the air or ocean. Every conversion shifts a slice of high-grade into low-grade, and the universe's total entropy (disorder) rises. Crucially, this is a one-way street . Heat flows from hot to cold on its own, but never

## Why no power station is ever 100% efficient

05 Why no power station is ever 100% efficient Put the two laws together and you get the headline result of this whole chapter. Any source that works by burning fuel to make heat, then converting heat to motion — coal, gas, oil, nuclear, even biomass — is a heat engine, and heat engines are hard-capped by the second law. You can only extract work from heat as it flows from a hot source to a cold sink, and a large fraction must be dumped to the cold side as waste. That waste isn't bad engineering; it's physics. Energy pathway Heat engine? Typical real ef

| Energy pathway                    | Heat engine? | Typical real efficiency | Why                                                       |
|-----------------------------------|--------------|-------------------------|-----------------------------------------------------------|
| Brown-coal electricity (Wirndook) | Yes          | ~25–33%                 | Wet, low-energy lignite + unavoidable heat-engine losses  |
| Combined-cycle gas                | Yes          | ~50–60%                 | Captures more of the heat in two stages, but still capped |
| Solar PV                          | No           | ~15–22%                 | Not a heat engine, but photons missed/lost as heat        |
| Hydro / wind                      | No           | ~80–90%                 | Converts existing motion directly — very few losses       |
| Solar hot water (direct)          | No           | ~60–70%                 | Sunlight → heat → hot water, almost no conversion steps   |

## Energy quality: match the tool to the job

06 Energy quality: match the tool to the job Because the second law makes high-grade energy precious, a smart energy choice matches the quality of the energy to the quality the job actually needs. Using premium electricity to do a job that only needs low-grade warmth is like using filtered drinking water to flush a toilet — it works, but you've wasted something valuable. High-grade · concentrated Save it for work Electricity, chemical fuels, compressed gas. Can be converted into almost any form. The most useful and most "expensive" in second-law terms —

## The implications — making the Tarra Cove choice

07 The implications — making the Tarra Cove choice This is the exact skill the study design names: turning the two laws into decisions. Every conversion step in a chain pays a second-law tax, and the taxes multiply. So one rule dominates: fewer conversions, and use the source closest to the form you need. SCENE · The conversion chain — pick a pathway, watch the losses compound Coal → grid → heater Gas → direct flame Solar thermal → tank 5 conversion steps 70 units wasted as heat 30 units of useful warmth The five practical implications you should be abl

## Which law is it? — sort the statements

### ◇ ACTIVE RECALL

08 Which law is it? — sort the statements The single most common exam slip is naming the wrong law. Tap a statement, then tap the bin it belongs in. First = about energy being conserved / where it went. Second = about quality, waste heat, entropy or direction. Both = needs the two laws together. First Law Second Law Both Laws s

## Decode the command word

### ◆ COMMAND WORDS

09 Decode the command word The verb in the question tells you how much to write. Tap each to see what the marker wants for a thermodynamics question.

## Six traps that cost real marks

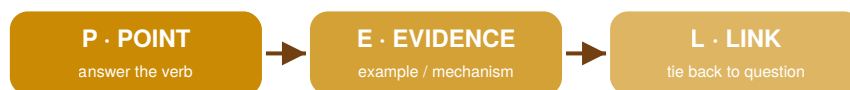
### ⚠ EXAM TRAPS

10 Six traps that cost real marks Each of these has burned students in real VCAA-style answers. Open them and steal the fix.

## P·E·L writing trainer

### 🔑 P·E·L WRITING

11 P·E·L writing trainer P oint → E xplain → L ink. Write a 3-mark answer below. As you type, the targets light up when you hit the ideas a marker is hunting for. This is live keyword detection — it rewards substance, not length. Explain, using both laws of thermodynamics, why a brown-coal power station can never convert all of its fuel energy into electricity. (3 marks) Aim for ~50–80 words. Name both laws, say what happens to the missing energy, and why it's unavoidable. 0/6 Start typing — name the first law to light your first target.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

## Same question, two answers

### ✗ WEAK ANSWER

weak answer and a

### ✓ STRONG ANSWER & MARKING

strong answer to the same 4-mark prompt. The highlights show exactly where marks are won and lost. Question. The Tarra Cove council is choosing between coal-fired electricity (then used in resistive heaters) and a direct solar hot-water system to warm water for its swimming pool. Using the laws of thermodynamics, justify which is the more efficient choice. (4 marks) ✗ Weak — 1/4 Solar is better because it's renewable and doesn't pollute . Coal is bad for the environment and wastes a lot of energy . The so



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

## Practice exam questions

### ★ PRACTICE & SELF-MARK

13 Practice exam questions / 20 marks Five VCAA-style questions, scaling from 2 to 6 marks. Write your answer first, reveal the model answer and marking guide, then award yourself stars. Your scores save automatically and feed the dock below. 0% Your KK06 mastery Award stars on each question to track how confident you are across all 20 marks. my scores

## One-screen cheat sheet

### ♦ RECAP / CHEAT SHEET

14 One-screen cheat sheet First Law — say it like this Energy is conserved — created/destroyed by nothing, only transformed. Inefficiency  $\neq$  loss; the energy became low-grade waste heat in the surroundings. Implication: no perpetual motion / over-unity machine can exist. Second Law — say it like this Every conversion increases entropy ; some energy degrades to dispersed heat. Energy quality always falls ; heat flows hot  $\rightarrow$  cold, never the reverse for free. Implication:  $\eta < 100\%$  always ; heat engines are capped low. Choice rules Fewer conversion steps = more

## ★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

### ✗ WEAK ANSWER

Solar is better because it's renewable and doesn't pollute . Coal is bad for the environment and wastes a lot of energy . The solar option is more efficient and cheaper so the council should pick it. Names no law. "Renewable / doesn't pollute" is a different dotpoint — it doesn't answer a thermodynamics question. "Wastes energy" is asserted, never explained. No mention of conversion steps or energy quality.

### ✓ STRONG / MODEL ANSWER

Heating water needs only low-grade thermal energy . The coal path makes electricity through many conversions (chemical  $\rightarrow$  heat  $\rightarrow$  motion  $\rightarrow$  electricity  $\rightarrow$  heat), and by the second law each step degrades some energy to waste heat , so only  $\sim 33\%$  of the coal's energy survives. The first law confirms the rest isn't destroyed — it's lost as heat to the surroundings. The solar system converts sunlight directly into heat in far fewer steps , so much less energy is degraded. Because it matches a low-grade need with a low-grade source , the solar option is the more efficient choice. Names both laws and uses them correctly, tracks the conversion chain, invokes energy quality matching, and lands a clear ju